Journal of Infection and Public Health 10 (2017) 836–841

Contents lists available at ScienceDirect

Journal of Infection and Public Health

journal homepage: http://www.elsevier.com/locate/jiph

Longitudinal study of Thai people media exposure, knowledge, and behavior on dengue fever prevention and control

Smith Boonchutima[a,*], Kirati Kachentawa[b], Manasanun Limpavithayakul[c], Anan Prachansri[c]

[a] *Chulalongkorn University, Thailand*
[b] *National Institute of Development Administration, Thailand*
[c] *Ministry of Public Health, Thailand*

ARTICLE INFO

Article history:
Received 12 August 2016
Received in revised form 23 January 2017
Accepted 28 January 2017

Keywords:
Dengue
Aedes aegypti
Thailand
Media
Knowledge
Behavior
Prevention
Control

ABSTRACT

Dengue hemorrhagic fever is transmitted through a bite by a dengue -infected *Aedes aegypti* mosquito. It was first reported in the mid -20th century in Thailand, and since then its epidemiology has been of great concern and has spread all across the country. The alarming incidence of dengue posed a serious threat to human health in all major cities of Thailand.

This study was aimed at identifying the level of awareness of dengue fever in Thai population knowledge for prevention and control, and most importantly contribution of media in educating masses for dengue control measures. It is longitudinal in nature and was conducted in 25 provinces of Thailand during 2013–2015. Approximately 7772 respondents participated in this study, with the selection of provinces based on considerations like population, prevalence and demography. A pre-tested structured questionnaire was used to collect information relevant to study participants' demographic profile, pre-existing knowledge about dengue fever and its reinforcement through media, and population attitudes toward prevention and control.

Over the period of three years, a positive trend was revealed relevant to the contribution of media in educating and reminding the Thai population of dengue, without any uniformity or powerful campaigns. Based on the results drawn from this study, we conclude that despite the measures undertaken to prevent dengue fever, there is insufficient media exposure. An interdisciplinary approach involving the community participation, media, and government is needed to overcome dengue threat in Thailand.

Introduction

Dengue hemorrhagic fever is caused by a virus transmitted by the mosquito *Aedes aegypti* causing significant morbidity and mortality all over the world. It has been estimated that almost 390 million individuals globally face this infection [1]. Currently, there is no effective vaccine or antiviral available to combat this infection except controlling the vector (mosquito) involved in the transmission of this infection [2]. As such, major attention has been paid to multi-pronged approaches involving integrated vector management strategies [3,4]. It has been recommended that countries prone to dengue spread should also have media campaigns tailored for dengue control through enhancing the community perception and behavior [4–6].

Thailand, a dengue-prone region, was reported in 2015 as having dengue infections that exceeded 140,000. The projected situation for the year 2016 is grim [7]. Since the very first discovery of dengue in Thailand in the year 1949, there has been several sporadic outbreaks off and on challenging the overall public health system of the country, however, the worst outbreak occurred in 1987 with 174,285 cases being reported [8]. The dengue fever is ranked as one of the most prevalent communicable diseases in Thailand and leading causes of sickness among children [9]. Since its first discovery, over the past six decades, the country has been struggling through various measures for overcoming this infection. A clinical trial involving 4000 Thai children between the ages of 4 and 11 were subjected to vaccine administration in the year 2012 in collaboration with Sanofi Pasteur, the manufacturer of this vaccine and promising results provided a hope for saving people from dengue, however, still its implementation is on the way [10–12].

* Corresponding author.
E-mail addresses: Smith.b@chula.ac.th, smith.boon@gmail.com (S. Boonchutima), Laterrepolka2011@gmail.com (K. Kachentawa), timdpc7@gmail.com (M. Limpavithayakul), anandp126@yahoo.co.th (A. Prachansri).

http://dx.doi.org/10.1016/j.jiph.2017.01.016
1876-0341/

In the absence of an effective antiviral and availability of good vaccine, mosquito (vector) control activities have been the primary method used to prevent the spread of the infection across the country. The vector control programs were carried out concurrently with health and educational activities for creating awareness among the population. Eight Thai ministries such as the education, and health joined hands to expand the awareness program in schools, ensuring that school going children have enough knowledge to protect them from dengue virus through avoiding mosquito bites. Such interventions used by Thailand government against dengue virus have been creating awareness through health education; getting rid of mosquito breeding sites; and the provision of immediate medical care. However, in spite of all these efforts dengue still poses a mega threat to the country [7].

Earlier days, the Thai Government National Plan for the Prevention and Control of Dengue Hemorrhagic Fever emphasized on the following measures for overcoming dengue issue [13].

i. Improvement of the environment at home and in the community in order to render it unfit for the breeding of *Aedes* mosquitoes.
ii. Human development — This will include education, information and communication activities aimed at raising the awareness of the people to be conscious about the danger of dengue and to prevent themselves from possible exposure to the disease. The plan will also emphasize the creation of a public norm of controlling the environment and destroying the breeding sites of *Aedes* mosquitoes.
iii. Development of technology to promote dengue prevention and medical care for dengue patients. This plan will emphasize measures to build people's capacity to take proper care of the sick persons at home and to build up responsive medical and health manpower to provide effective treatment for dengue patients.

Infectious disease experts also suggested a community-based approach to the prevention and control of dengue in Thailand and other countries [14]. Relatively, this approach is cost-effective and most efficient intervention in the long term. Community-Based Empowerment Programs (CBEP) in the past had been used across Thailand to increase society participation through collaboration. The CBEP involved members of the community coming together and brainstorming. These programs were meant to enhance the learning experience among the members of a community, foster a better understanding of dengue and coming up with collaborative efforts in eradicating the dengue virus in the community. The CBEPs from different zones carried out monthly meetings to discuss the findings and the way forward. The effectiveness of the community-based programs was evaluated regularly with conclusive thoughts that these endeavors had made some strides in eliminating the virus, but still needed more community participation and involvement of other stakeholders to make the programs more efficient [15].

Several factors contribute to the spread of the dengue virus, which have made the preventive measures difficult to implement or rather insufficient to contain the virus. There has been a societal change as a result of population increase in Thailand in the past couple of decades; constant movement of the populace, especially in the capital, Bangkok, and other major cities. Urbanization in Thailand has been uncontrollable leading to scarcity of resources, especially in the major cities leading to poor water management system that has increased the breeding sites of mosquitoes. The rural and also some urban areas are still struggling with the problem of piped water, causing the residents to store water in containers which are the main breeding site for *A. aegypti* mosquitoes. The mass movement of the population in different regions in Thailand has led to the transfer of the virus from high endemic areas to low endemic areas.

Among several other strategies, knowledge about the dengue-spreading mosquito and its control is a critical factor in controlling the infection [5,16–18]. Furthermore, a reiteration of certain facts like dengue control measures and methodologies through media are supposed to be effective methodologies. In the backdrop of lingering dengue issue in Thailand, we firmly believe that knowledge imparted through the media relevant to dengue control will significantly help to overcome the infection. As such, this specific study was designed to evaluate the public basic knowledge relevant to dengue and media role in providing information through reiterative messages associated with dengue. Our study provides important information relevant to strengthening dengue control measures knowledge through continued media campaigns.

Methods

This longitudinal study utilized a descriptive cross-sectional research design. The study was initiated in January 2013 and concluded in December 2015. The study was approved by the Research Ethics Committee of the Department of Disease Control. No identifier or respondent information was stored and hence declared exempt. To participate in this study, volunteers must be fifteen years of age or above and have no record of mental illness. Before administering the questionnaire, a study description written at the layman level was provided to consenting individuals. Upon reading, the individuals willing to participate were asked to fill up a questionnaire. This well-structured questionnaire collected data about the general perception of dengue fever and awareness programs on the media relevant to prevention and control from 25 provinces of Thailand. The 24 provinces were selected with two provinces each from the 12 Offices of Disease Control, which are branches of the Department of Disease Control, selected based on the highest and lowest prevalence rate. The 25th province selected in the sample was Bangkok since it is the Capital city of Thailand. Throughout the research, different respondents were used to collect data, with volunteers taking approximately 10 min to fill the questionnaire. Four sections of the questionnaire used to collect information include:

i. Demographic profile.
ii. Knowledge about dengue fever.
iii. Awareness about prevention.
iv. Control measures.

Questions relevant to demographic profile include gender, age, education level, occupation, and income. Concerning media exposure, we asked the participants to provide information relevant to their experience in watching the programs relevant to dengue in five-time frames including, every day, more than three times per week, once a week, once a month or never saw the dengue relevant programs on the TV. The media exposure, in this context, was interpreted as the frequency of an individual to get exposed to the message related to dengue in the media or other means of communication. This follows a section on three questions in True/False setup relevant to mosquito breeding, dengue disease, and preventive measures. The logical flow of questionnaire culminated as a part of the action by the study participants associated with destroying breeding sites of the mosquito responsible for transmitting dengue.

Results

This longitudinal study spans over three years, starting in the year 2013 and concluding in 2015. Participants of the study were

Table 1
Study participants demographic profile.

	Year 2015 (n = 2607)		Year 2014 (n = 2842)		Year 2013 (n = 2323)	
	n	%	n	%	n	%
A. Gender						
Male	1280	49.1	1329	47.5	1158	49.8
Female	1327	50.9	1493	52.2	1165	50.2
B. Age						
Age (Year 2014–2015)						
15–24	496	19.0	623	21.9		
25–34	523	20.1	665	23.4		
35–44	575	22.1	614	21.6		
45–54	466	17.9	406	14.3		
55–64	330	12.7	283	10.0		
65+	217	8.3	251	8.8		
Age (Year 2013)						
15–30					851	36.6
31–45					450	19.4
46–60					656	28.2
60+					366	15.8
C. Education						
Education level						
Without education or lower than primary education	41	1.6	64	2.3	97	4.2
Primary education	984	34.3	790	27.8	436	37.0
Secondary education/vocational certificate	914	35.1	990	34.8	664	28.6
Diploma/ high vocational certificate	262	10.0	347	12.2	196	8.4
Bachelor's degree	409	15.7	536	18.9	392	16.9
Higher than Bachelor's degree	74	2.8	84	3.0	63	2.7
Others	9	0.3	17	0.5	4	0.2
D. Occupation						
Occupation						
Agriculture	471	18.1	450	15.8	522	22.5
Freelance	490	18.8	518	18.2	446	19.2
House husband/wife	176	6.8	183	6.4	155	6.7
Business/self employed	537	20.6	557	19.6	478	20.6
Student	336	12.9	358	12.6	211	9.1
Unemployed	122	4.7	87	3.1	66	2.8
Gov't officer/state officer	371	14.2	501	17.6	280	12.1
Industrial staff	69	2.6	120	4.2	91	3.9
Others	27	1.0	55	1.9	43	1.9
E. Monthly income						
Monthly income						
Below 2500 Baht	234	9.0	497	17.5	357	15.4
2,501–10,000 Baht	1267	48.6	1361	47.9	1122	48.3
10,001–20,000 Baht	434	16.6	600	21.1	385	16.5
Above 20,000 Baht	246	9.4	263	9.3	263	11.4

randomly through accidental sampling. The study was completely described to each respondent before his/her consent to fill in the questionnaire. We collected responses from a total of 7772 individuals, 2323 in the year 2013, 2842 the year 2014 and 2607 in 2015. As far as the gender ratio is concerned, the relative proportion of females was a bit higher when compared with male participants (Table 1). The mean age in years 2015, 2014 and 2013 are 41, 39 and 43 respectively.

It is important to mention here that in the year 2013, we grouped study participant age wise using a 15 years age range, whereas for the years 2014/15 this we used a ten years range. In all these three years, the majority of the individuals consenting to participate in the study have their education level as primary or secondary (60%). As far as the occupations are concerned, the majority of respondents had their own business or self-employed. Their overall proportion was 20.6% in 2015, 19.6% in 2014 and 20.6% in 2013, followed by freelancers' group at 18.8% in 2015, 18.2% in 2014 and 19.2% in 2013. Approximately half of the participating individuals' income ranges between 2501–10,000 Baht (Thai currency) representing the middle–low-income household. The conversion rate of Thai baht against the US dollar is about 1 to 35.

Media plays a major role in the dissemination of public health knowledge [16,19,20]. Particularly, dengue control is associated with the knowledge about the overall life cycle of virus-spreading mosquitoes and precautionary safety measures for avoiding mosquito bites including destroying breeding sites. We evaluated Thai mass media contribution toward dengue awareness. Respondents were asked to get their thoughts recorded in five categories, as far as media contribution is concerned. Relevant to media exposure study participants were asked whether they had the opportunity to see dengue-related information in the media i) Everyday, ii) More than 3 times per week, iii) once a week, iv) once a month, and v) never. In the year 2013, the media was recognized for spreading dengue knowledge once per month category receiving the highest affirmative response (32.4%) and never for 4%. The situation improved in the year 2014 with more than three times per week category emerged as the highest with 35.2% with never 5.1%. The third year of this study, 2015, showed a differential pattern than the previous two, at least once a week category with 33.4% emerged on the top, followed by once a month (33.1%) and the lowest one never for 3%.

There are several disease/infections that can be prevented through precautionary measures and in this respect, knowledge about the disease-causing agents is helpful to stop the transmission of disease. In this study, knowledge relevant to the dengue virus spreading vector *A. aegypti* mosquito was emphasized by Thailand's Department of Disease Control in the hope that it would help the population to avoid mosquitoes through various strategies, thus saving themselves from acquiring dengue virus infection.

Table 2
Change in dengue relevant general knowledge of Thai over the years (2013–2015).

Items knowledge	N	%
2013		
Aedes aegypti mosquito lays eggs in dirty smelly water (False)	779	33.5
Symptoms of dengue fever in the early stage include high fever, lethargy, and anorexia (True)	2136	92.0
To effectively eradicate *Aedes aegypti* mosquito, we must destroy their larvae weekly (True)	2213	95.3
Average percentage of respondents who answered the questions correctly		**73.6**
2014		
Only village health volunteers and public health officers that are responsible for preventing dengue (False)	1892	66.6
Symptoms of dengue fever in the early stage include high fever, lethargy, and anorexia and if you have one, you must see the doctor (True)	2356	82.9
To effectively eradicate *Aedes aegypti* mosquito, we must destroy their larvae weekly (True)	2701	95.0
Average percentage of respondents who answered the questions correctly		**81.5**
2015		
Water containers can be mosquito breeding sites (False)	2513	96.4
All people should eradicate the larvae by themselves (True)	2469	94.7
Aedes aegypti mosquito lays eggs in dirty smelly water (False)	782	30.0
Average percentage of respondents who answered the questions correctly		**73.3**

Note: The percentage shown represents the percentage of people who answered correctly.

In the knowledge domain, we asked the respondents to mark our questionnaire for three qualifying statements with Yes/No option and the respondents have to manifest his/her knowledge in one of these. These statements and the number of respondents who gave correct answers are provided in Table 2. In response to the first question in Table 2, "*Aedes aegypti* mosquito lay eggs in the dirty water surface" 33.5% of the responses were correct in 2013 and 30% correct in 2015. The correct response increased in the second knowledge question, "symptoms of dengue fever in the early stage include high fever, lethargy, and anorexia" with 92% and 82.9% of the respondents showing the right response in 2013 and 2014 respectively. In the third question, "to effectively eradicate mosquitoes that can spread dengue, we should destroy their larvae weekly" there was an anomalous response, whereby 95.3% and 95% of the respondents provided correct responses in 2013 and 2014 correspondingly. In 2014, only 66.6% of the respondents gave correct responses for the knowledge question "only village health volunteers and public health officers are responsible for preventing dengue (False)". In 2015, 96.4% of the respondents gave appropriate answers to the first knowledge question "water containers can be mosquito breeding sites (False)". Approximately 94.7% of respondents indicated that all people should eradicate larvae by themselves. There is no appropriate explanation why the percentage of respondents who showed correct responses about the knowledge question, "symptoms of dengue fever in the early stage include high fever, lethargy, and anorexia" decreased from 92% in 2013 to 82.9% in 2014. All in all, further longitudinal research may help elaborate the anomaly exemplified in this knowledge question. The sampled respondents displayed encouraging findings concerning knowledge of breeding sites. Nonetheless, they did not have specific knowledge on the type of water to supports breeding of *A. aegypti* mosquitoes. It is highly likely that the sampled population confused malaria and dengue control, hence the variation in the responses. It is worthwhile noting that a significant number of the respondents in 2015 understood that it is their responsibility to eradicate *A. aegypti* mosquitoes breeding sites.

Based on the knowledge, we also asked, the response of the study participants for implementing mosquito control measures for the year 2014 and 2015. It is encouraging to note that the number of people paying attention to the mosquito control has increased from 28 to 35% in less than once a week category (Table 3).

Since this study is a longitudinal one and the sample collection involves differential conditions as well as the research teams performing the work, we performed repeated measures analysis of variance (ANOVA) to evaluate differences between the means of 2013, 2014, and 2015 with regards to media exposure, knowledge on dengue fever and behavior on dengue prevention and control. One sample t-test was used for finding the mean difference of the years 2014 and 2015. Findings from Table 4 revealed that the respondents in 2015 were more exposed to the media ($\bar{x} = 3.37$) compared with 2014 ($\bar{x} = 3.31$) and 2013 ($\bar{x} = 2.84$). Taking into account each variable separately the respondents from 2015 had the highest frequency in the media exposure ($\bar{x} = 3.37$, P-value = 0.001) and manifested the highest recommended behaviors for mosquito control ($\bar{x} = 4.24$, P-value = 0.001). The respondents from 2014 have the precise knowledge regarding the prevention and control techniques of dengue fever ($\bar{x} = 0.82$, P-value = 0.001) compared to the other years (Table 4).

Discussion

The causative agent for dengue fever is dengue virus transmitted from one person to another by the *A. aegypti* mosquito bite. There is no effective vaccine or antiviral for controlling dengue fever. One of the best control measures presently is the controlling breeding sites of mosquitoes. However, the public needs to have a good understanding of the disease and mosquito breeding site. Our study provides important information relevant to the media efforts in educating public relevant to dengue and basic knowledge among the Thai people relevant to dengue control. Besides public health knowledge, it has been recognized that dengue infection is multifactorial in nature. Sporadic outbreaks manifest several environmental factors associated with the mosquito development and capability to transmit disease.

In the year 2013, one-third of the respondents (32.4%) had an opportunity to see dengue control relevant programs once a month. This number reduced in the two subsequent years 2014 and 2015 to 19.4 and 17.2%, respectively demonstrating a major shift as far as media efforts over three years of this longitudinal study is concerned. In comparison with the developed nation's media programs relevant to dengue, the trend is not so encouraging. The low percentage can be attributed to a large number of households having earned less than 10,000 Baht or a large number of individuals who have reached the secondary level of education and lower. The results, on the other hand, show that 42% of the respondents were exposed to the media either more than 3 times per week or even daily which is enough to create awareness, but considering that everyone is at risk of the dengue fever, the level of awareness and exposure should be near 100%. The situation in 2014 in terms of media exposure is less or more similar to that of 2013 with approximately 51% of the respondents being exposed to the media once a week, less than once a week and even some not exposed at all: this

Table 3
Behaviors for dengue fever prevention and control in 2014–2015.

How often do you eradicate or clean mosquito breeding sites, such as big water jars, trays, vases, flower pot trays, and water puddles?	Year 2015		Year 2014	
	N	%	N	%
Once a week	1260	48.3	1738	61.2
Less than once a week	918	35.2	796	28.0
Never	240	9.2	180	6.0
Without potential mosquito breeding sites	183	7.0	133	4.7

Table 4
Comparison between the exposure to media messages, knowledge, and behavior regarding the prevention and control techniques of dengue fever in 2013–2015.

	Means	S.D.	F	P-value
Media exposure				
Year 2013 (n = 2319)	2.84	1.145	158.161	<0.001
Year 2014 (n = 2842)	3.31	1.084		
Year 2015 (n = 2607)	3.37	1.013		
Knowledge				
Year 2013 (n = 2306)	0.14	0.213	6942.650	<0.001
Year 2014 (n = 2842)	0.82	0.237		
Year 2015 (n = 2607)	0.74	0.192		
Behavior				
Year 2014 (N = 2842)	3.87	1.513	137.463	<0.001
Year 2015 (N = 2607)	4.24	0.904	239.637	<0.001

Remarks: Exposure to media messages and behaviors have a maximum score of 5, knowledge has a maximum score of 1.

shows that a significant amount of the population is not adequately exposed to the media to gain the necessary information on dengue fever. In 2015, 53% were exposed to the media once per month, less than once per month or never, this shows a consistent trend in lack of exposure for a large population.

In the year 2013, 73.6% of the respondents were aware of the dengue fever prevention and control measures. Despite such a huge percentage being aware of the measures, the results show that a lesser percentage applies the knowledge. Secondly, the huge level awareness cannot be directly attributed to media exposure because most of them might have accessed the information through community programs/initiatives or other sources of information such as friends and books/journals. The disparity in the level of knowledge and media exposure is a clear indication that there are other means that the individuals can access information. Still, the media plays a huge part in conveying the information especially bearing the fact that it has a wider reach, and holds higher perceived credibility than many other means.

In 2014, there was an increase in the level of knowledge about the prevention measures and control as 81.5% answered the questions correctly. One of the reasons that might have led to the increase of this number is the 2013 dengue fever outbreak in Thailand that might have led to the increase in awareness creation which in turn saw more than 47% of the respondents having media exposure for 2–4 times per week or even more. The other factor that might have led to the increase would be a difference between the respondents between the years. In 2015, the level of knowledge fell from 81.5% to 73.7% in 2014. Perhaps, the reason for this fall is the decrease in media exposure which occurred at a recovery period after the 2013 outbreak, which may have seen the reluctance in awareness by the government and other relevant bodies. The second reason for this fall might be low memory/information retention capacity in the population or lack of consistency.

In 2015, 48.3% of the respondents practiced the correct method of eradicating mosquito breeding sites which is supposed to be done once a week for the process to be effective. The percentage of 'without potential mosquito breeding sites' that decreased obviously compare to 2014. This may be due to increased awareness of breeding sites because of the outbreaks so they spent efforts to look for breeding sites. In 2014, there was no outbreak so they did not look for the mosquito breeding sites.

In 2014, 61.2% of the respondents were aware of the correct method. The main reason can be the presence of the 2013 dengue fever, which created a high alert going into 2014, but there was reluctance after the outbreak and the recovery stage. It is notable that in Thailand the concern about dengue fever is not consistent, but only applied by many when there is imminent danger.

Recommendations

Based on data generated from our study, the need for an effective, and consistent media campaign is quite clear. It is important to mention that people are getting away from conventional media like TVs, radio and getting more engaged in social media. Differential responses relevant to dengue program on the media could be the individual's habits being diverted to Facebook and other social media. Further studies will be needed to clarify this issue, and if this is the case, then social media should be actively involved in the dengue control campaign. Furthermore, the urge for previously implemented program exists in the country. Media exposure plays a critical part in creating awareness due to its wide reach and relatively economic nature. Media exposure is limited in certain cases like low-income households who cannot afford the common sources of information, illiterate people who cannot comprehend the information, and children who are too young to listen and apply this information. Community programs are critical in complementing the efforts of media as it can better serve areas that media fail to cover. Community participation can be invoked through showing concern, initiating dialogue, creating community ownership and health education.

An inter-provincial coordination should be adopted in the Thailand scenario by recognizing the social, economic, and environmental problems that promote mosquito breeding. The prevention and control of dengue can be complex for the government to handle it alone, therefore; it is imperative to have inter-organizational coordination which involves the pooling of resources together. It is not enough for the National Dengue Prevention and Control Committee to formulate a national plan in fighting against dengue fever, but it is on the committee to implement the plan through proper planning, directing, monitoring, and review. The program needs to be continuously monitored to ensure that parts are implemented according to standards, and there should be continuous improvement. Children, poor people and low-income earners, are at a higher risk because of their vulnerability, ignorance, or even helplessness. It is upon the government and other stakeholders to adopt a more aggressive and hands-on approach to help these vulnerable groups. There should be constant inspections and evaluations of the environment and such task should not be left in the hands of the community alone, but it should be a collaborative effort from all stakeholders linked to the community. Without community participation, the media campaign simply aiming to increase individuals' awareness proved not effective to control the spread of dengue.

Limitations

We would like to particularly mention limitations of our study. To be eligible for this study, the participants should know "How to read and write". As such, the illiterate and mentally unstable people were automatically excluded. Essentially, for dengue control campaign every citizen, either illiterate or literate should be aware of the dengue prevention and control measures.

Funding

No funding sources.

Competing interests

None declared.

Ethical approval

Not required.

References

[1] Wenceslao Orellano P, Reynoso JI, Stahl HC, Salomon OD. Cost-utility analysis of dengue vaccination in a country with heterogeneous risk of dengue transmission. Vaccine 2016;34(5):616–21.
[2] Ndii MZ, Allingham D, Hickson RI, Glass K. The effect of Wolbachia on dengue outbreaks when dengue is repeatedly introduced. Theor Popul Biol 2016;111:9–15.
[3] Hapuarachchi HC, Koo C, Rajarethinam J, Chong CS, Lin C, Yap G, et al. Epidemic resurgence of dengue fever in Singapore in 2013-2014: A virological and entomological perspective. BMC Infect Dis 2016;16(1):300.
[4] Harburguer L, Licastro S, Masuh H, Zerba E. Biological and chemical characterization of a new larvicide ovitrap made of plastic with pyriproxyfen incorporated for *Aedes aegypti* (Diptera: Culicidae) control. J Med Entomol 2016;53(3):647–52, http://dx.doi.org/10.1093/jme/tjw022.
[5] Nazareth T, Teodosio R, Porto G, Goncalves L, Seixas G, Silva AC, et al. Strengthening the perception-assessment tools for dengue prevention: a cross-sectional survey in a temperate region (Madeira, Portugal). BMC Public Health 2014;14:39.
[6] Schaffner F, Medlock JM, Van Bortel W. Public health significance of invasive mosquitoes in Europe. Clin Microbiol Infect 2013;19(8):685–92.
[7] Thailand braces for 'mega-outbreak' of dengue fever. Channel NewsAsia; 2016 [Accessed 7 November 2016] Available from: http://www.channelnewsasia.com/news/asiapacific/thailand-braces-for-mega /2465766.html.
[8] Limkittikul K, Brett J, L'Azou M. Epidemiological trends of dengue disease in Thailand (2000-2011): a systematic literature review. PLoS Negl Trop Dis 2014;8(11):e3241.
[9] Chao DL, Halstead SB, Halloran ME, Longini Jr IM. Controlling dengue with vaccines in Thailand. PLoS Negl Trop Dis 2012;6(10):e1876.
[10] Hadinegoro SR, Arredondo-Garcia JL, Capeding MR, Deseda C, Chotpitayasunondh T, Dietze R, et al. Efficacy and long-term safety of a dengue vaccine in regions of endemic disease. N Engl J Med 2015;373(13):1195–206.
[11] L'Azou M, Moureau A, Sarti E, Nealon J, Zambrano B, Wartel TA, et al. Symptomatic dengue in children in 10 Asian and Latin American countries. N Engl J Med 2016;374(12):1155–66.
[12] Pitisuttithum P, Bouckenooghe A. The first licensed dengue vaccine: an important tool for integrated preventive strategies against dengue virus infection. Expert Rev Vaccines 2016;15(7):795–8.
[13] Rojanapithayakorn W. Dengue haemorrhagic fever in Thailand. Dengue Bull 1998;22:60–72.
[14] Gubler DJ, Clark GG. Dengue/dengue hemorrhagic fever: the emergence of a global health problem. Emerg Infect Dis 1995;1(2):55–7.
[15] Therawiwat M, Fungladda W, Kaewkungwal J, Imamee N, Steckler A. Community-based approach for prevention and control of dengue hemorrhagic fever in Kanchanaburi Province, Thailand. Southeast Asian J Trop Med Public Health 2005;36(6):1439–49.
[16] Doblecki-Lewis S, Chang A, Jiddou-Yaldoo R, Tomashek KM, Stanek D, Anil L, et al. Knowledge, attitudes, and practices of Florida physicians regarding dengue before and after an educational intervention. BMC Med Educ 2016;16(1):124.
[17] Hairi F, Ong CH, Suhaimi A, Tsung TW, bin Anis Ahmad MA, Sundaraj C, et al. A knowledge, attitude and practices (KAP) study on dengue among selected rural communities in the Kuala Kangsar district. Asia Pac J Public Health 2003;15(1):37–43.
[18] Swaddiwudhipong W, Lerdlukanavonge P, Khumklam P, Koonchote S, Nguntra P, Chaovakiratipong C. A survey of knowledge, attitude and practice of the prevention of dengue hemorrhagic fever in an urban community of Thailand. Southeast Asian J Trop Med Public Health 1992;23(2):207–11.
[19] Arellano C, Castro L, Diaz-Caravantes RE, Ernst KC, Hayden M, Reyes-Castro P. Knowledge and beliefs about dengue transmission and their relationship with prevention practices in Hermosillo, Sonora. Front Public Health 2015;3:142.
[20] Ramzan M, Ansar A, Nadeem S. Dengue epidemics: knowledge perhaps is the only key to success. J Ayub Med Coll Abbottabad 2015;27(2):402–6.